

1. Research Lifecycle

- a. **Problem Identification:** The phase of identifying a gap in current technology or a real-world problem that requires a computational solution.
- b. **Literature Review:** Analyzing *state-of-the-art* methodologies from previous research to understand existing limitations and ensure the proposed solution is a fresh solution.
- c. **Formulation:** Defining specific research questions and establishing measurable hypotheses to address the identified problem.
- d. **Methodology Design:** Designing the system architecture, selecting appropriate datasets, and defining evaluation metrics (e.g., utilizing the LANL Cyber Security Dataset and designing a TGN architecture with PSO).
- e. **Implementation & Experimentation:** Writing the programmatic code, building the computational model, and executing systematic testing scenarios.
- f. **Evaluation & Analysis:** Measuring and interpreting the experimental results against the defined metrics (such as evaluating the AUROC and Recall scores of the anomaly detection model).
- g. **Scientific Publication:** Compiling the research findings, methodology, and conclusions into a standardized academic format (IEEE) for global peer review.

2. Explanation of Differences

- a. **Research Problem:** The broad, underlying issue or operational challenge (high false alarm rates in network anomaly detection due to extreme class imbalance).
- b. **Research Question:** The specific, targeted inquiry the research aims to directly answer.
- c. **Hypothesis:** A testable, predictive statement regarding the relationship between the experimental variables.
- d. **Experiment:** The systematic computational procedure conducted to validate or invalidate the hypothesis.

3. Experimental Variables

- a. **Control Variable:** The factors kept strictly constant to ensure a fair and isolated test (utilizing the exact same TGN core architecture and LANL dataset across all tests).
- b. **Independent Variable:** The specific functional factor systematically manipulated by the researcher (the threshold calibration method: static threshold vs. dynamic PSO optimization).
- c. **Dependent Variable:** The measured outcome or performance metric that changes in response to the manipulation (the final AUROC and anomaly Recall scores).